

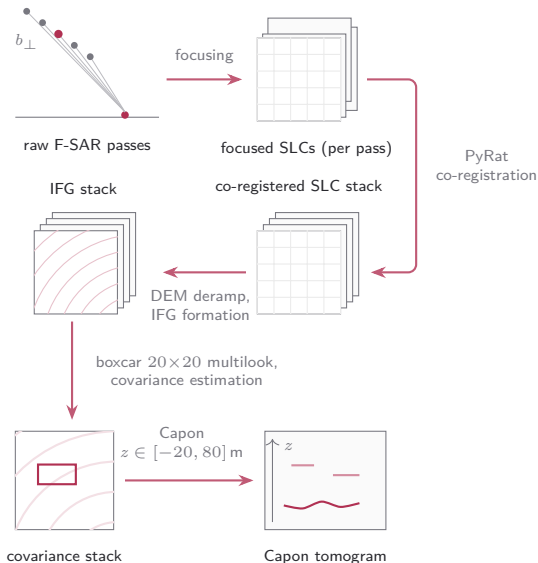
DLR-TomoSAR — Dataset & ground-truth labels

F-SAR stack · prominence-ranked fitting · best- K selection · GPU-scale extraction

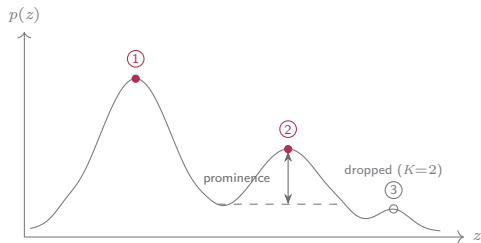
July 2026

F-SAR stack and preprocessing

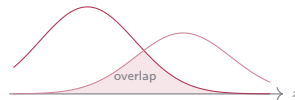
- Airborne F-SAR stack over an urban test site; primary PS02, secondaries selected at training time (PS04/06/08/26) — the **reduced stack**.
- Per-track acquisition geometry extracted from campaign metadata: per-pixel baselines, look angles, slant ranges (consumed by the physics losses, Act VI).
- Every artifact is written self-describing: config, stats and geometry ride with the data.



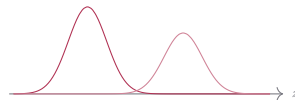
Before the fit: prominence-ranked initialisation



Raw Capon spectrum; peaks ranked by prominence, top K kept.



wide σ start: overlapping components, coupled gradients



small σ_0 start: disjoint components, easier convergence

- Candidate scatterers: local maxima with prominence $\geq 5\%$ of the pixel maximum, separated by at least the initial width σ_0 .
- Ranked by prominence, top K kept — μ_k , a_k read directly at the surviving peaks; every component starts at the shared width guess $\sigma_0 = \max(2 \Delta\xi, \text{span}/8K)$, with $\Delta\xi$ the elevation grid step and $\text{span} = \xi_N - \xi_1$ the full grid extent (100 m here): an eighth of a component's even share of the axis, floored at two grid bins — deliberately small, so components start disjoint and their gradients decouple (right).
- Degenerate pixels handled explicitly: fewer than K peaks $\rightarrow$ argmax of the peak-masked residual; empty profile $\rightarrow$ uniform spread over the grid.

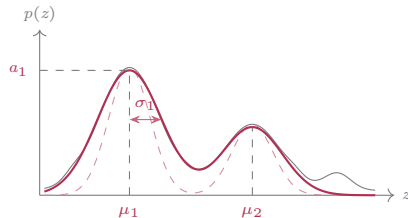
Ground truth: fitted Gaussian parameters

Supervised target: the per-pixel array $[a_k, \mu_k, \sigma_k]_{k=1}^K$ ($K=2$ today, $K=5$ in development), fit to the per-pixel max-normalised Capon spectrum $\tilde{p} = p/p_{\max}$ on the elevation grid ξ_n :

$$\mathcal{L} = \frac{1}{N} \sum_{n=1}^N (\hat{p}(\xi_n) - \tilde{p}(\xi_n))^2, \quad \hat{p}(\xi) = \sum_k a_k e^{-\frac{(\xi - \mu_k)^2}{2\sigma_k^2}}$$

One shared loss; each parameter group takes its own projected Adam step:

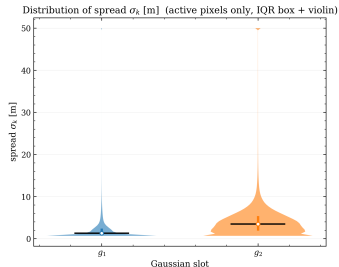
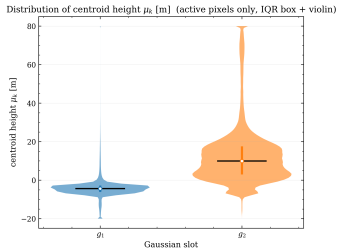
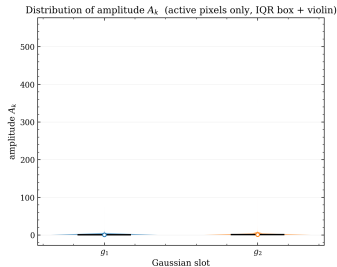
$$\begin{aligned} a_k &\leftarrow [a_k - \eta \text{Adam}(\partial_{a_k} \mathcal{L})]_+ \\ \mu_k &\leftarrow \text{clip}_{[\mu^-, \mu^+]} (\mu_k - \eta \text{Adam}(\partial_{\mu_k} \mathcal{L})) \\ \sigma_k &\leftarrow \text{clip}_{[\sigma^-, \sigma^+]} (\sigma_k - \eta \text{Adam}(\partial_{\sigma_k} \mathcal{L})) \end{aligned}$$



Same pixel: Capon spectrum, **peak initialisation (dashed)**, **converged fit**; the fitted $[a_k, \mu_k, \sigma_k]$ are the label.

- Fitted in normalised space; the winning amplitudes are denormalised back, $a_k \rightarrow p_{\max} a_k$ (μ_k, σ_k already live on the physical axis).
- All pixels batched in one optimisation — not a per-pixel loop (next slide); most pixels activate one or two Gaussians (Act III).

What the labels look like: fitted parameter distributions



- Distributions of the supervised targets $[a_k, \mu_k, \sigma_k]$ over **active pixels only**, split by slot (g_1, g_2 ; IQR box + violin).
- The second slot g_2 sits **higher** ($\mu_2 \approx 10$ m vs $\mu_1 \approx -5$ m) and **broader** (σ_2 heavier-tailed) — it takes the elevated, more diffuse scatterer when a pixel activates two.
- Amplitudes stay small and tightly peaked; μ and σ carry the long tails the robust scaling (previous act) is built for.

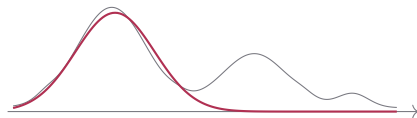
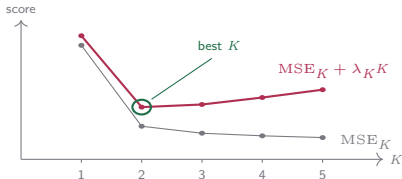
Choosing K : fit every order, penalise complexity

- Every active pixel is fitted at every order $K = 1, \dots, K_{\max}$, all sharing the same peak init (top- K slice).
- Per-pixel selection by penalised score

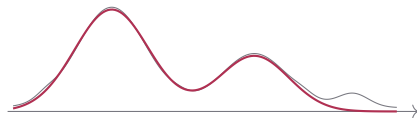
$$S(K) = \underbrace{\text{MSE}_K}_{\text{fitting component}} + \underbrace{\lambda_K K}_{\text{regularization component}}$$

per- K MSE, penalised scores and the best- K map ride with the dataset as diagnostics.

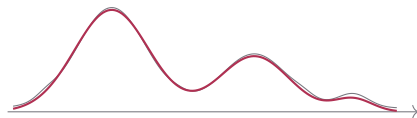
- Today's $K=2$ labels run with $\lambda_K = 0$: the fitting component alone decides — the regularization component becomes load-bearing for the $K=5$ extension.



$K=1$: second scatterer missed

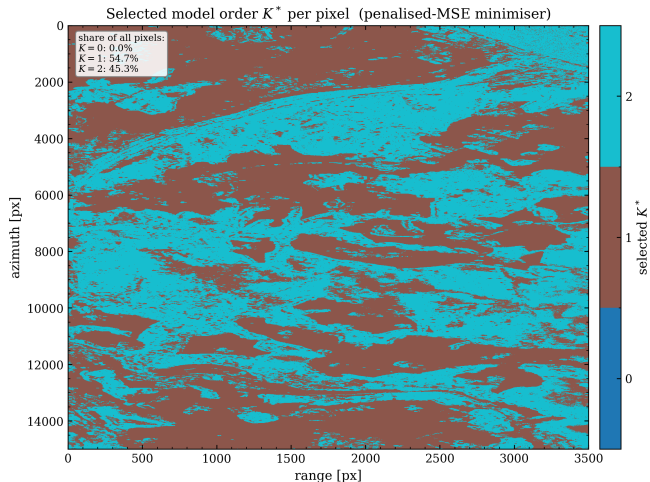


$K=2$: both scatterers captured



$K=3$: marginal MSE gain

The best- K map: where the second scatterer lives



- The penalised-MSE selection of the previous slide, run per pixel over the full scene.
- $K=1$: **54.7%**, $K=2$: **45.3%**, $K=0$: 0.0% — the “most pixels activate one or two Gaussians” claim, now measured.
- The two orders track **spatial structure**: contiguous regions of single- vs double-scatterer response, following the scene's layover geometry — not per-pixel noise.
- Rides with the dataset as a diagnostic; the complexity price λ_K becomes load-bearing for the $K=5$ extension.

Label generation at scale: multi-GPU parameter extraction

- The fit reformulated as batched gradient optimisation: vectorised over all pixels, sharded across GPUs — full-scene extraction in ~ 4 minutes.
- Labels stop being precious: sweeps over K , regularisation and fit modes are routine; datasets are regenerated, never patched.

